Building a Planet: Plate Boundaries in Action

Tectonic Explorer · Concord Consortium

Year 8	**~30 minutes**	**Science · Earth and Space Science**

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This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://tectonic-explorer.concord.org/?planetWizard=true
Curriculum links: AC9S8U03 · AC9S8U04

Learning intentions

- I can describe what happens at convergent, divergent and transform plate boundaries.
- I can link specific landforms (mountains, rifts, volcanoes) to the type of plate boundary that created them.
- I can explain how the rock cycle connects to tectonic processes over long timescales.

Getting started

Open Tectonic Explorer and use the planet wizard to set up your own planet — choose how many continents to place and where. Once your planet is built, run the simulation forward, and use the cross-section view to see what is happening beneath the surface.

Tasks

1. Before you run the simulation, predict what you think will happen where two continents drift toward each other. Write your prediction down.

2. Run your planet forward until two plates collide. Describe what you see forming in the cross-section view at the collision point.

3. Find an example of a convergent, a divergent and a transform boundary in your model. Record what forms at each one.

Boundary type	What you observed forming

4. Choose one convergent boundary in your model. Use the cross-section view to explain what is happening to the rock at the point where one plate goes underneath the other (subduction).

5. Find a divergent boundary in your simulation. What landform is starting to appear where the plates are pulling apart?

6. Scientists once thought continents were fixed in place. Using what you've observed in this simulation, explain what evidence tectonic activity provides for the idea that Earth's plates move over time.

7. Rocks formed at a divergent boundary (like new ocean floor) are geologically 'young' compared to rocks in the middle of a continent. Using your understanding of the rock cycle, suggest why this might be true.

8. If you were designing a city on your simulated planet, which type of plate boundary would you avoid building near, and why? Refer to at least one specific hazard.

Extension (optional)

Real Earth has transform boundaries (like the San Andreas Fault) where plates slide past each other without creating or destroying crust. Set one up in your model and explain why this type of boundary produces earthquakes but not usually mountains or volcanoes.